

DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

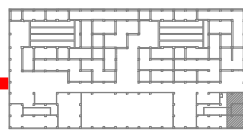
APOL-504

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	515	3FA	80A-APOL-ST3-N-6532-01	1 OF 1	1	CJ-M15W14-PR-A20-000149	2A	
2	515	3FA	80A-APOL-ST3-N-6533-01	1 OF 1	1	CJ-M15W14-PR-A20-000149	2A	
3	515	3FA	80A-APOL-ST3-N-6534-01	1 OF 1	1	CJ-M15W14-PR-A20-000149	2A	
4	515	3FA	80A-APOL-ST3-N-6535-01	1 OF 1	1	CJ-M15W14-PR-A20-000149	2A	
5	515	3FA	100A-APOL-ST3-N-6503-01	1 OF 1	1	CJ-M15W14-PR-A20-000148	1A	
6	515	3FA	100A-APOL-ST3-N-6504-01	1 OF 1	1	CJ-M15W14-PR-A20-000148	1A	
7	515	3FA	100A-APOL-ST3-N-6505-01	1 OF 1	1	CJ-M15W14-PR-A20-000148	1A	
8	515	3FA	100A-APOL-ST3-N-6506-01	1 OF 1	1	CJ-M15W14-PR-A20-000148	1A	
9	515	3FA	200A-APOL-ST3-N-6501-01	1 OF 1	1	CJ-M15W14-PR-A20-000148	1A	
10	515	3FA	200A-APOL-ST3-N-6502-01	1 OF 1	1	CJ-M15W14-PR-A20-000148/149	1A/2A	
11	515	3FA	200A-APOL-ST3-N-6531-01	1 OF 1	1	CJ-M15W14-PR-A20-000149	2A	

MAKER PACKAGE	
ITEM NO.	W14-PD-4603 A
SERVICE	A-POLYMER AUTO DISSOLVING UNIT
Q'TY	1
SIZE	W x L x H m
CAPA.	kg/hr
MAT'L	STS304
POWER	kw

대외비

APOL-504



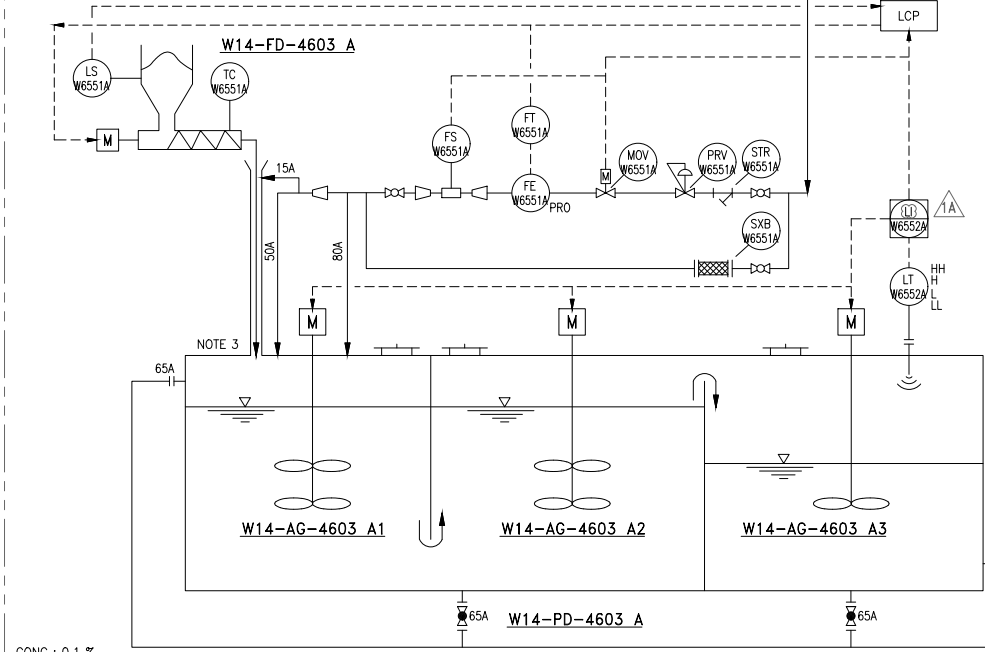
504 KEY PLAN (3F)

1. 약품종류에 따라 LINE 색상을 구분하여야 한다.
2. MER 용해장치 CURB 및 PUMP AREA에서 발생되는 물은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 배수된다.
3. 승강기 PLATFORM, HANDRAIL 설치 할 것.
4. AGITATOR MAT'L 표기 : SHAFT/BLADE
5. 시 CLOSE 되는 내관에는 LOTO 설치.
6. INTERNAL PSV SET PRESSURE : 0.65 Mpag

539

INDUSTRIAL WATER
M15W13-PR-A20-00081-A
FROM
UDD FOR INDUSTRIAL WATER

A-POLYMER AUTO DISSOLVING UNIT MAKER PACKAGE

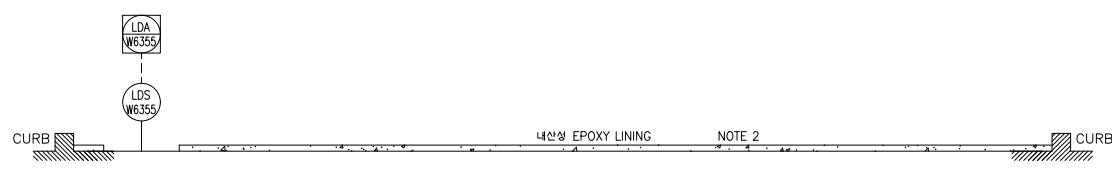


516

APOL-504

504

A-POLYMER
M15W14-PR-A20-000149
FROM A-POLYMER DISSOLVING UNIT
W14-PD-4603 B

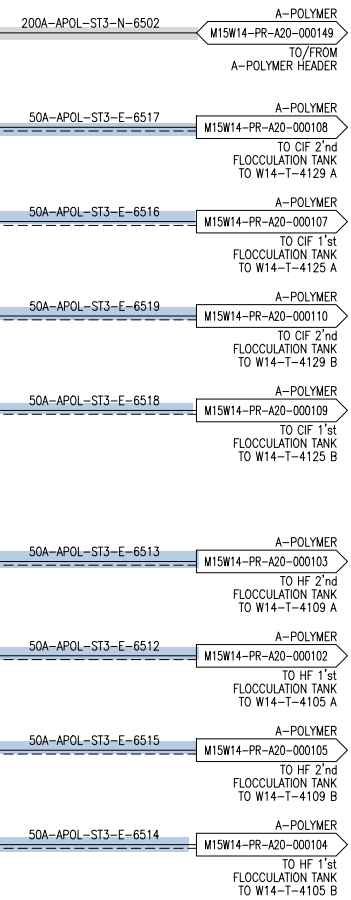
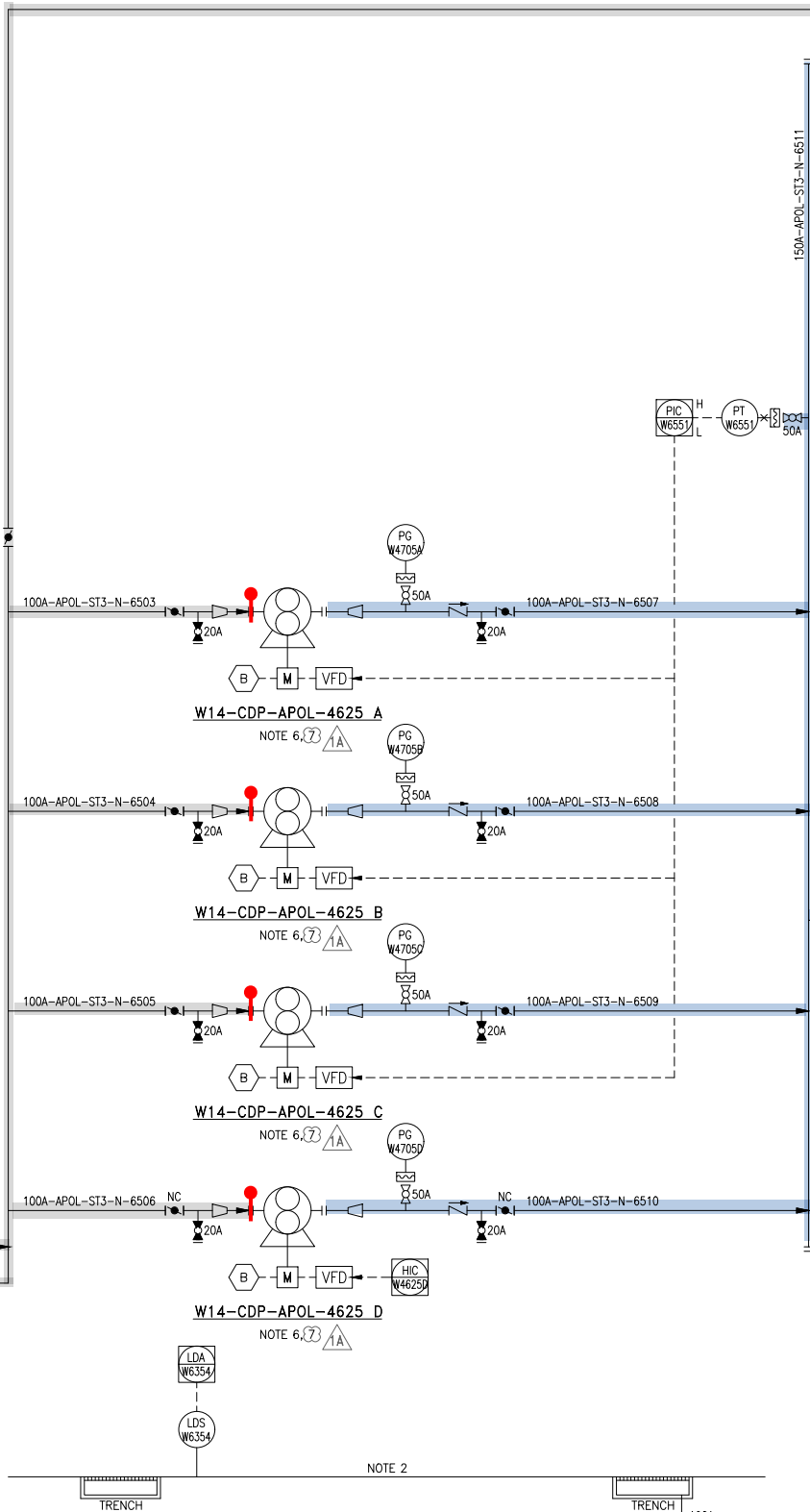


A-POLYMER AUTO DISSOLVING UNIT MAKER PACKAGE : DESIGN BY SUPPLIER

ITEM NO.	W14-FD-4603 A
SERVICE	A-POLYMER FEEDER
Q'TY	1
TYPE	SCREW FEEDER
CAPA.	kg/hr
MAT'L	STS304
POWER	kw

ITEM NO.	W14-AG-4603 A1/A2/A3
SERVICE	A-POLYMER AGITATOR
Q'TY	3
TYPE	PROPELLER
CAPA.	1,175
SPEED	rpm
MAT'L	STS304L/STS304L
POWER	kw

ITEM NO.	W14-CDP-APOL-4625 A/B/C/D
SERVICE	A-POLYMER FEED PUMP
Q'TY	3 + 1
TYPE	ROTARY LOBE (VVF)
CAPA.	m3/hr.ea x MPa
MAT'L	(NBR/GC250/SCM440)
POWER	kw



1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30

수정	실명	날짜
REV.	DESCRIPTION	DATE

제도	이인학
DRAWN BY	

검토	한진희
CHECKED BY	

승인	박동명
APPROVED BY	

제작사	SK 에코플랜트
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공사명	M15X Ph4 WWT Project
PROJECT TITLE	

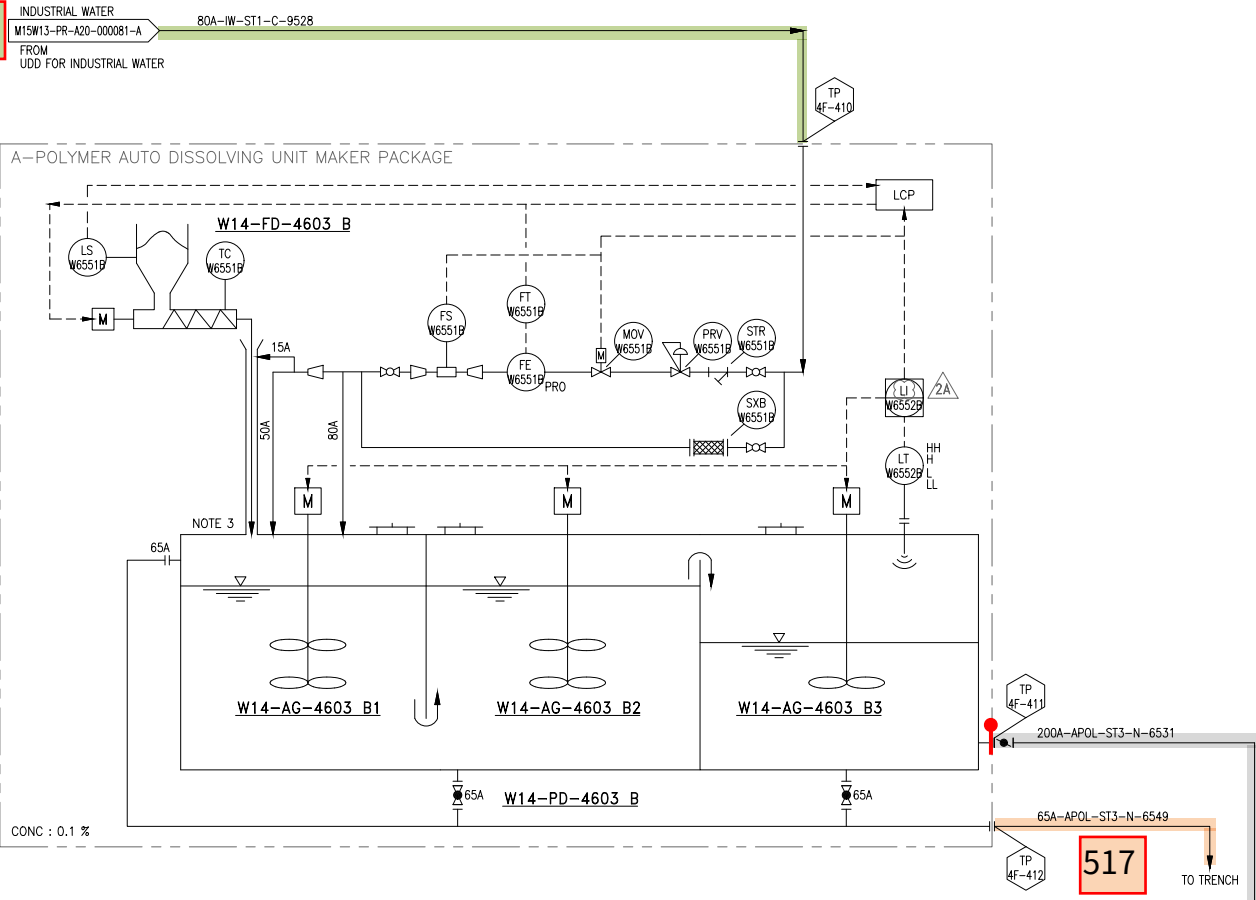
도면명	PIPING & INSTRUMENT DIAGRAM
DWG. TITLE	

축척	NONE
SCALE	

날짜	2025. 08. 14
DATE	

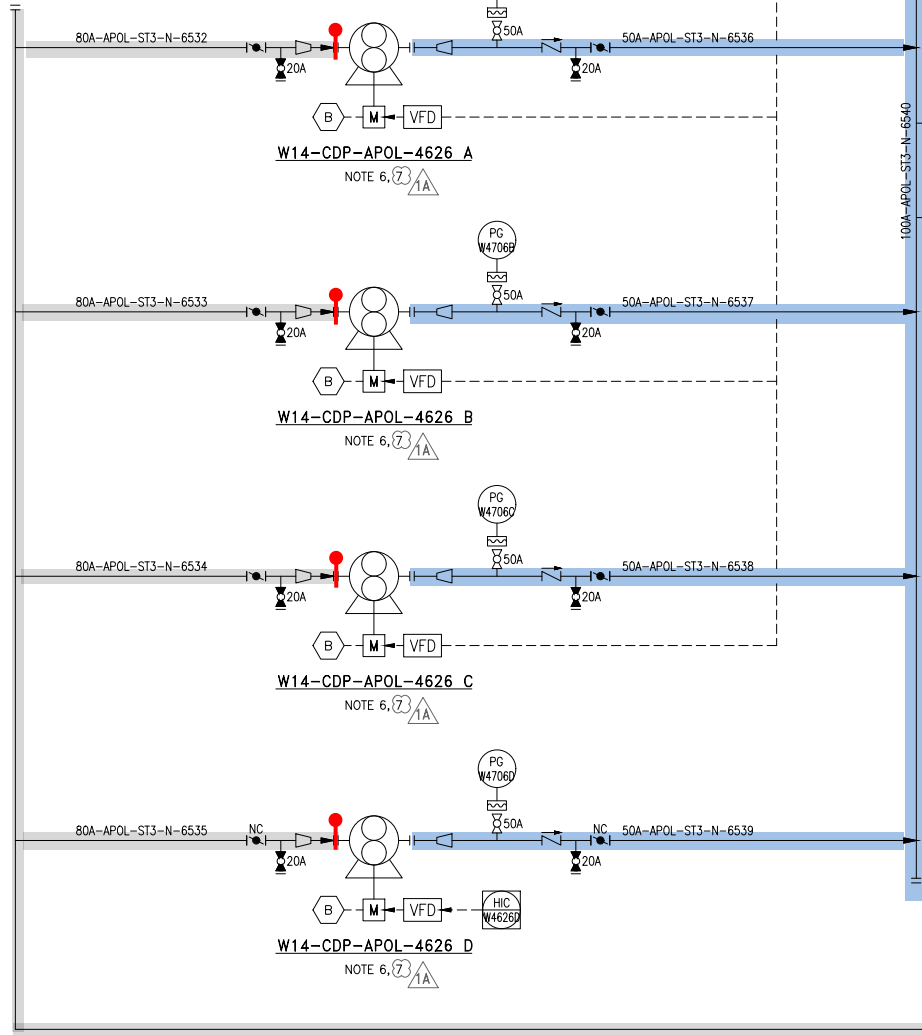
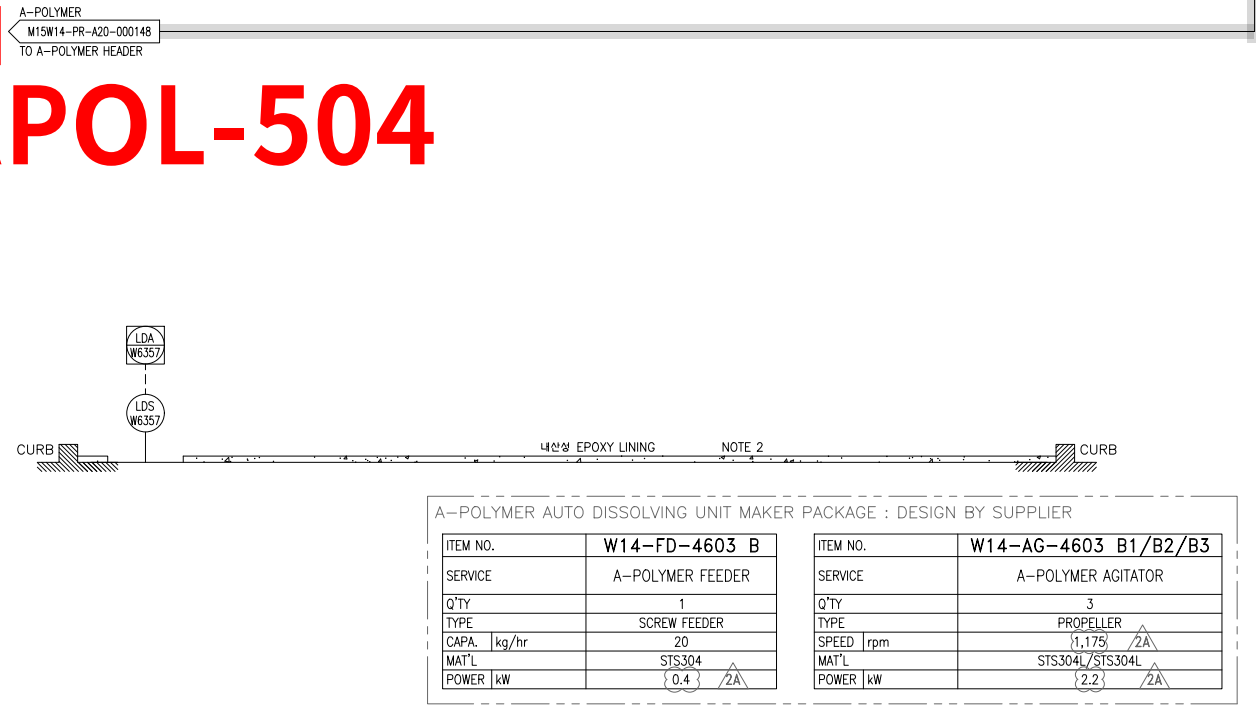
PROJECT NO.	22330D	REV.	A
도면번호	SHEET NO. CJ-M15W14-PR-A20-000148	1A	

539



504

APOL-504



APOL-504

대외비

KEY PLAN (3F)

NOTE

1.약품종류에 따라 LINE 색상을 구분하여야 한다.

2.POLYMER 용해장치 CURB 및 PUMP AREA에서 발생되는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.

3.장부 PLATFORM, HANDRAIL 설치 할 것.

4.AGITATOR MAT'L 표기 : SHAFT/BLADE

5.본인 시 CLOSE 되는 배관에는 LOTO 설치.

6.INTERNAL PSV 설치 할 것.

7.INTERNAL PSV SET PRESSURE : 0.65 Mpag

443

A-POLYMER
M15W14-PR-A20-000120
TO WW
FLOCCULATION TANK
TO W14-T-4194 A

A-POLYMER
M15W14-PR-A20-000121
TO WW
FLOCCULATION TANK
TO W14-T-4194 B

443

A-POLYMER
M15W14-PR-A20-000502
TO BRINE 1st
FLOCCULATION TANK
W14-T-4724

A-POLYMER
M15W14-PR-A20-000503
TO BRINE 2nd
FLOCCULATION TANK
W14-T-4728

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이인학
DRAWN BY	검	토한진희
인	인	박동영
DVD BY	제	작사
504	SK 에코플랜트	
공	사	명
PROJECT TITLE	M15X Ph4 WWT Project	
도면명	DWG. TITLE	
PIPING & INSTRUMENT DIAGRAM		
A-POLYMER AUTO DISSOLVING UNIT & FEED PUMP (2/2)		
속	척	
SCALE	NONE	
날	짜	
DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호	SHEET NO. CJ-M15W14-PR-A20-000149	
2A		

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1

OR  **SK ecoplant**



LL COUPLING : FC * HALF COUPLING : HF/HF1

1. (*) FIELD ADJUST

[illegible]



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			



SK ecoplant



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* SUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* SUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			



SK ecoplant



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* SUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* SUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			



4

SK ecoplant



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME
	THICK.	-	DESIGN	1.0	60	UNIT 3FA								M15X Ph4 Project
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-030008-A	 1	25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY APOL	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000148	 0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. 100A-APOL-ST3-N-6503	22330D
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	SH'T NO. 01 / 01	REV. 1

고 : 그 기밀은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 기타 무단 이용을 금지하며, 회사 내 모든 포자의 폐기 절차에 따라 폐기하여야 하며, 위와 같은 위반행위 발생 시 손해배상 청구에 응할 수 있는



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* SUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* SUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT

6

CONTRACTOR 

COMMODITY	APOL
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LINE NO.	100A-APOL-ST3-N-6504
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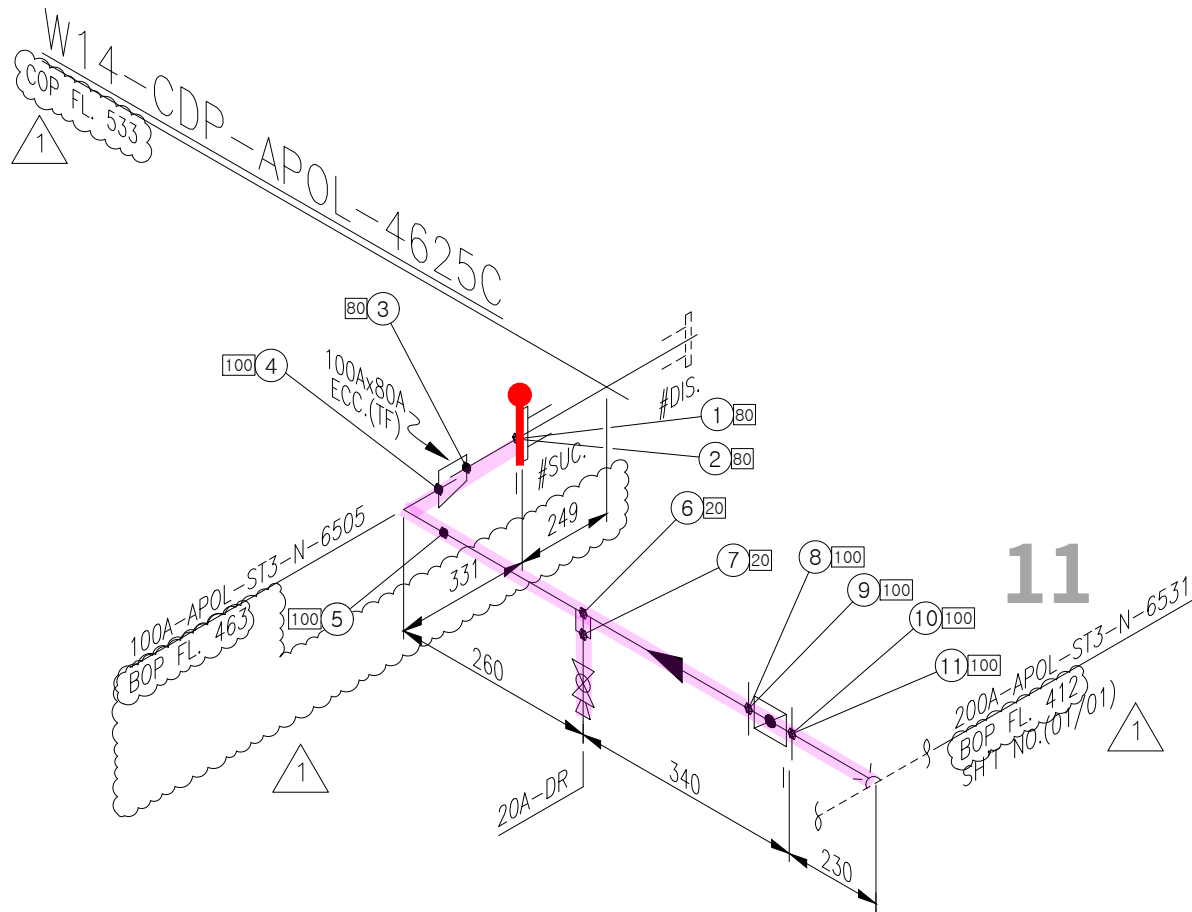
CONTRACTOR	PROJECT NAME
	M15X Ph4 Project

COMMODITY	APOL	JOB NO.
		22330D

LINE NO.	100A-APOL-ST3-N-6504	SH'T NO.	REV.
		01 / 01	1



ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	100A	0.5M	PIPE, STS 304TP, ERW, BE – SCH 10S
P	80A	0.1M	PIPE, STS 304TP, ERW, BE – SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
E90	100A	1EA	ELBOW 90 DEG LR, STS304, ERW, BE – SCH 10S
RE	100Ax80A	1EA	REDUCER ECC, STS304, ERW, BE – SCH 10S
HF	20A	1EA	COUPLING HALF, STSF304, SW – SCH 10S
PLG	20A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
VF	100A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-GCD450 /
			TRIM-STS304 / EPDM-LINED, LEVER
F	100A	2EA	FLG SO, KS 10K, RF, STS304
F	80A	1EA	FLG SO, KS 10K, RF, STS304
G	80A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	80A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	80A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
BL	100A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (*C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	-	DESIGN	1.0	60	UNIT 3FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-030008-A	 1	25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY APOL	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000148	 0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. 100A-APOL-ST3-N-6505	22330D	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		SH'T NO. 01 / 01	REV. 1

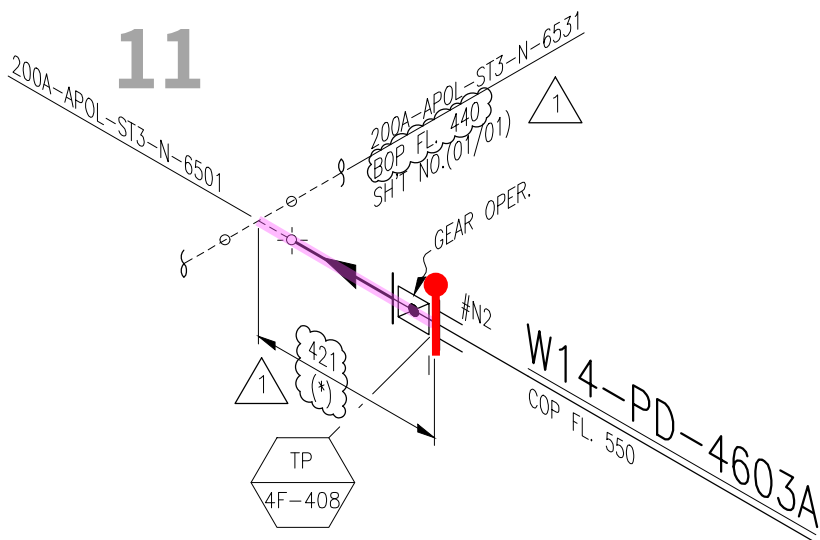
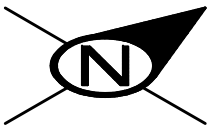


* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* SUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* SUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			



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TOR  SK ecoplant



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT	3FA							M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-030008-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	APOL	22330D
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.
						CJ-M15W14-PR-A20-000148								200A-APOL-ST3-N-6501	01 / 01
															REV. 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

[illegible]

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB1L0/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT	3FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-030008-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	APOL	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000149								200A-APOL-ST3-N-6502	01 / 01	1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

